Examiner only

**6.** The cell surface membranes of root cells are responsible for differences in the concentration of ions in the roots when compared to those found in the surrounding soil solution. Inorganic ions are essential for cell metabolism in plants.

*(a)* Describe the role of magnesium, calcium, and phosphate ions in plant cells and tissues. [3]

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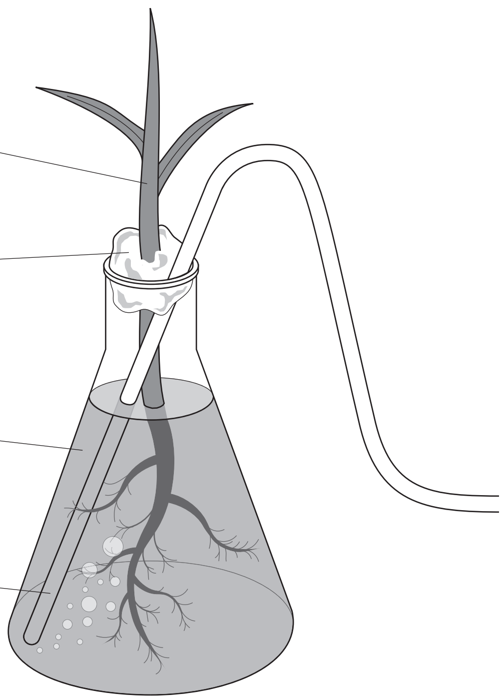
To investigate the uptake of mineral ions by plants, barley seedlings were grown with their roots in a solution of mineral ions for four days. Air was bubbled through the solution. The concentration of the ions in the solution and the root tissue were determined.




| ion | ion concentration / mmol $dm^{-3}$ | |
|---|---|---|
| | surrounding solution | root tissue |
| calcium ($Ca^{2+}$) | 1.0 | 57.0 |
| magnesium ($Mg^{2+}$) | 0.3 | 33.0 |
| phosphate ($PO_4^{3-}$) | 0.9 | 1.6 |

Examiner only

(b) (i) With reference to the data, explain the relative concentrations in the surrounding solution and the root tissue after four days. [3]

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(ii) Explain why the growth of a barley seedling may be reduced if the solution did not have air bubbled through it. [4]

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